

Yelp Restaurant Business Performance Analysis Report

1. Title Page

Yelp Restaurant Business Performance Analysis

Business Intelligence Dashboard Report

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Course: Bachelor of Computer Applications (BCA)

Project Type: Data Analytics / Business Intelligence Project

Tools Used: Python, MySQL, Excel Dashboard

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2. Executive Summary

This report presents a comprehensive analysis of restaurant business performance using the Yelp dataset. The objective of this project is to help stakeholders understand the restaurant industry before launching a new restaurant business.

A dashboard was developed to analyze restaurant market performance, customer engagement patterns, seasonal trends, rating distribution, and top-performing brands. This analysis provides valuable business insights that can support strategic decision-making.

The dashboard focuses on identifying:

- Best cities for restaurant business expansion
- Peak customer traffic hours
- Customer rating patterns
- Seasonal business trends
- Top restaurant brands for collaboration opportunities

The insights derived from this analysis help stakeholders make informed, data-driven business decisions.

3. Problem Statement

A leading business group plans to enter the restaurant industry and wants to understand the restaurant market before making strategic investment decisions.

To support this decision-making process, a Restaurant Business Performance Dashboard was developed using Yelp dataset analysis.

The dashboard focuses on:

- Top-performing cities for restaurant business expansion
- Restaurant traffic during peak and low engagement hours
- Distribution of restaurant ratings
- Seasonal business performance trends
- Top restaurant brands for potential collaboration or benchmarking

This analysis helps stakeholders better understand market opportunities and challenges before entering the restaurant industry.

4. Dataset Overview

The dataset used in this project is the Yelp Open Dataset.

Main tables used:

- Business
- Review
- Tip
- Check-in
- User

Filtered Dataset: Only restaurant-related businesses were selected using category filtering.

Total records after cleaning:

- Total Restaurants: 46,945
 - Total Brands: 32,498
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5. Tools & Technologies

Tools used in this project:

- Python
- Pandas
- MySQL
- Excel
- SQL
- Data Visualization

Libraries:

- Pandas
 - SQLAlchemy
 - PyMySQL
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6. Data Cleaning & Processing

The raw Yelp dataset contained multiple business categories and large volumes of records.

Data cleaning steps performed:

- Removed null values
- Filtered only restaurant businesses
- Cleaned duplicate brand names
- Processed check-in timestamps
- Extracted hourly engagement patterns
- Created cleaned restaurant-specific tables

Restaurant tables created:

- restaurant_business
- restaurant_review
- restaurant_tip
- restaurant_checkin

This improved analysis quality and reduced noise.

7. KPI Metrics

The dashboard contains four major KPI cards:

Total Restaurants

Total number of restaurant businesses in the dataset.

Value: 46,945

Total Brands

Total unique restaurant brands.

Value: 32,498

Average Rating

Average rating of all restaurants.

Value: 3.47

Average Review Count

Average review count per restaurant.

Value: 45.95

8. Dashboard Analysis

8.1 Top Cities by Success Score

Cities were analyzed using a success score based on:

- Average Rating

- Review Count
- Restaurant Count

Top-performing cities:

- Reno
- New Orleans
- Boise

Low-performing city:

- Edmonton

Observation: Reno shows the highest restaurant business potential.

8.2 Restaurant Traffic by Check-in Hour

Check-in data was analyzed to identify customer traffic patterns across 24 hours.

Peak Traffic: 5 PM – 12 AM

Lowest Traffic: 7 AM – 10 AM

Observation: Restaurant activity is highest during evening and late-night hours.

8.3 Rating Distribution

Restaurant ratings were analyzed to understand customer satisfaction patterns.

Most restaurants rated: 3.5 – 4.0 stars

Least restaurants rated: 1.0 – 1.5 stars

Observation: Most restaurants operate in the average-to-good rating range.

8.4 Seasonal Trends

Seasonal engagement was analyzed across:

- Summer
- Winter
- Spring
- Fall

Observation:

- Summer shows highest engagement
- Fall shows lowest engagement

Seasonal variation exists but is moderate.

8.5 Top Brand Performance

Restaurant brands were analyzed using performance score.

Top-performing brands:

- McDonald's
- Chick-fil-A
- Chipotle Mexican Grill

Observation: Large established brands dominate customer engagement and performance.

9. Key Insights

Key insights from the analysis:

- Restaurant market is highly competitive with 46K+ restaurants.
 - Reno is the strongest city for restaurant business opportunities.
 - Peak customer traffic occurs during evening and late-night hours.
 - Most restaurants receive ratings between 3.5 and 4.0.
 - Summer shows slightly stronger business performance.
 - Large brands dominate market engagement.
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10. Business Recommendations

Based on dashboard analysis, the following recommendations are suggested:

City Selection

Focus on high-performing cities like Reno for expansion.

Operating Hours

Prioritize evening and late-night business operations.

Customer Experience

Improve service quality to maintain ratings above 4.0.

Seasonal Strategy

Prepare seasonal campaigns for high-demand periods.

Brand Partnerships

Consider collaborations with strong-performing restaurant brands.

11. Conclusion

The Yelp Restaurant Business Performance Dashboard provides valuable insights into restaurant market performance, customer behavior, and business opportunities.

This analysis helps stakeholders understand:

- Where to expand
- When demand is highest
- Which brands perform best
- How customer satisfaction is distributed

The dashboard enables data-driven decision-making and reduces business risk before entering the restaurant industry.

Overall, this project demonstrates the practical use of data analytics, SQL, and business intelligence in solving real-world business problems.
